



CYBERSECURITY

# Cybersecurity Fundamentals and Threats

Understanding the foundations of digital defense — core concepts, the modern threat landscape, and how organizations protect themselves.

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*An Introduction for Teams, Students & New Analysts*

# What We'll Cover

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# What Is Cybersecurity?

**Cybersecurity is the practice of protecting systems, networks, devices, and data from unauthorized access, disruption, or damage.**

It combines technology, processes, and people to reduce risk and keep information confidential, accurate, and available when it's needed.



**Protect systems**



**Secure networks**



**Defend data**



**Reduce risk**



**Goal**

**Keep the right people  
in — and everyone  
else out.**

*Security is never “done” — it's a continuous cycle of assessing risk, defending assets, and responding to new threats.*

# The CIA Triad

Every security control ultimately serves one of these three goals.



## Confidentiality

Only authorized users can access sensitive information.



## Integrity

Data stays accurate and unaltered from creation to use.



## Availability

Systems and data are accessible whenever legitimately needed.

# Why Cybersecurity Matters

**\$4M+**

Average global cost of a  
data breach, per industry reports

**1 every  
few sec.**

Estimated frequency of  
cyberattacks worldwide

**Majority**

Of breaches involve a  
human element — error or deception

*Beyond direct cost, breaches damage customer trust, halt operations, and can trigger regulatory penalties — making prevention far cheaper than recovery.*

# Common Threat Actors



## Cybercriminals

Financially motivated groups running scams, fraud, and ransomware for profit.



## Nation-State Actors

State-sponsored groups pursuing espionage, sabotage, or geopolitical advantage.



## Hacktivists

Ideologically driven attackers targeting organizations to make a political statement.



## Insiders

Employees or contractors who misuse legitimate access, intentionally or not.



## Script Kiddies

Less-skilled attackers using pre-built tools, often opportunistically.



## Organized Crime

Structured criminal groups running large-scale, coordinated cyber operations.

# Malware Types



## Virus

Attaches to files and spreads when the infected file runs.



## Worm

Self-replicates and spreads across networks with no user action.



## Trojan

Disguises itself as legitimate software to gain access.



## Ransomware

Encrypts data and demands payment for its release.



## Spyware

Secretly monitors activity and harvests sensitive data.



## Rootkit

Hides deep in a system to maintain undetected, persistent access.

# Phishing & Social Engineering



## Definition

Social engineering manipulates people into revealing information or taking actions that compromise security.

Phishing is its most common form — fraudulent messages that impersonate a trusted sender to steal credentials or deliver malware.



## Spear Phishing

Highly targeted attacks aimed at a specific person or role.



## Vishing

Voice-call scams that pressure victims to act quickly.



## Smishing

Phishing delivered through SMS or messaging apps.



## Pretexting

A fabricated scenario used to earn trust and extract data.

# Network-Based Attacks



## Man-in-the-Middle

An attacker secretly intercepts and possibly alters communication between two parties.



## DDoS

A flood of traffic from many sources overwhelms a system, taking it offline.



## DNS Spoofing

Corrupted DNS records redirect users to malicious, look-alike sites.



## Packet Sniffing

Unencrypted traffic on a network is captured and read by an attacker.

# Password & Authentication Attacks



## Brute Force

Systematically tries every possible password combination.



## Dictionary Attack

Tests common words and known passwords against an account.



## Credential Stuffing

Reuses passwords leaked from other breaches to break in.



## Password Spraying

Tries one common password across many accounts to avoid lockouts.

# Insider Threats

Not every threat comes from outside the organization. Insiders already hold trusted access.



## Malicious Insiders

Employees or contractors who intentionally steal data, sabotage systems, or abuse access for personal gain.

- Selling data to competitors
- Sabotage after termination
- Unauthorized system access



## Negligent Insiders

Well-meaning users whose careless mistakes create openings for attackers, without any intent to cause harm.

- Falling for phishing emails
- Weak or reused passwords
- Misconfigured cloud storage

# Anatomy of a Cyber Attack

Most attacks follow a similar chain of stages — breaking any link stops the attack.

## Reconnaissance

Research the target



## Delivery

Send it to the target



## Installation

Plant persistent malware



## Actions on Objective

Steal, damage, or disrupt



Build the exploit

## Weaponization



Trigger the vulnerability

## Exploitation



Establish remote access

## Command & Control

# Defense in Depth

No single control stops every attack. Layering defenses means one failure doesn't mean total compromise.



- **Policies & People**  
Security awareness, training, and clear procedures
- **Physical**  
Locked facilities, badges, and controlled access
- **Perimeter**  
Firewalls and gateway filtering at the network edge
- **Network**  
Segmentation, monitoring, and intrusion detection
- **Endpoint**  
Antivirus, device hardening, and patch management
- **Application & Data**  
Encryption, access controls, and secure coding

# Cybersecurity Best Practices



Use strong, unique passwords with multi-factor authentication



Keep software and systems patched and up to date



Back up critical data regularly, and test restoring it



Run ongoing security awareness training for all staff



Apply least-privilege access — grant only what's needed



Encrypt sensitive data at rest and in transit



Maintain a tested incident response plan



Segment networks to limit how far an attacker can move

## KEY TAKEAWAYS

# Security Is Everyone's Responsibility



### Think in layers

No single tool or rule stops every threat — defense in depth matters.



### People are the front line

Most breaches involve human error or manipulation, not just technical flaws.



### Prevention beats recovery

Proactive habits are far cheaper than responding to a breach after the fact.

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## Thank You

*Questions & Discussion*